

Review

Coverage of Exam

Roughly Chapter 5 to 15

Exactly:

Chapter 6 without 6.5, 6.6

Chapter 7 only 7.2

Chapter 8 without 8.5, 8.6

Chapter 9 only 9.2

Without Chapter 10

Chapter 11 without 11.4 11.6 11.7

Chapter 12 without 12.6

Chapter 13 without 13.5, 13.6

Chapter 14 without 14.9, 14.10

Chapter 15 without 15.8

Information of Exam

主要为解答题，有定义题（不超过十分）和判断题，不会超过课后题难度

Chapter 6 Applications of Definite Integrals

Note 6.1 Volumes Using Cross-Sections

Definition of Cross-Section

Computing the volume of a solid through integral of cross-section

$$\text{area}(V = \int_a^b A(x) dx)$$

Some examples of computing the solids of revolution(disk method and washer method)

Note 6.2 Volumes Using Cylindrical Shells

Strategy of slicing with cylinders

The shell method formula for revolution about a vertical line L

$$V = \int_a^b 2\pi(x - x_L) f(x) dx$$

Note 6.3 Arc Length

Definition of arc length and its formula ($L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$)

Formula for the form $x = g(y)$

Another form for arc length (differential form $L = \int ds$ where

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Note 6.4 Areas of Surfaces of Revolution

Definition and formula of surface area generated by revolving

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Formula for the form $x = g(y)$

Chapter 7 Integrals and Transcendental Functions

Note 7.2 Exponential Change and Separable Differential Equations

Definition and solution of differential equations ($\frac{dy}{dx} = f(x, y)$, solution is $y = y(x)$)

Definition and solution of separable differential equations

($\frac{dy}{dx} = g(x)H(y)$, solution is $\int h(y) dy = \int g(x) dx$ where $h(y) = \frac{1}{H(y)}$)

Chapter 8 Techniques of Integration

Note 8.1 Integration by Parts

Definition of integration by parts(

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx)$$

Reduction formula(recursive formula)

Integration by parts for definite integrals

Note 8.2 Trigonometric Integrals

Strategy for the integrals of the form $\int \sin^m x \cos^n x dx$

Strategy for the integrals of powers of $\tan x$ and $\sec x$

Strategy for the integrals of the form

$\int \sin mx \sin nx dx, \int \sin mx \cos nx dx, \int \cos mx \cos nx dx$ (transform the products into the sum)

Note 8.3 Trigonometric Substitutions

Cases for trigonometric substitution

- For $\sqrt{a^2 + x^2}$:
Let $x = a \tan \theta$, then $a^2 + x^2 = a^2 \sec^2 \theta$.
- For $\sqrt{a^2 - x^2}$:
Let $x = \sin \theta$ or $\cos \theta$, then $a^2 - x^2 = a^2 \cos^2 \theta$ ($a^2 \sin^2 \theta$).
- For $\sqrt{x^2 - a^2}$:
Let $x = \sec \theta$, then $x^2 - a^2 = a^2 \tan^2 \theta$.

Note 8.4 Integration of Rational Functions by Partial Fractions

Definition of partial fraction(rational function to a sum of simpler fractions)

Method of partial fractions(note the difference between the factor with degree one and two)

Heaviside Method

Chapter 9 First-Order Differential Equations

Note 9.2 First-Order Linear Equations

Definition and form of first-order linear equations($\frac{dy}{dx} + P(x)y = Q(x)$)

Method of solving first-order linear equations(multiply both sides by $v(x) = e^{\int P(x) dx}$ and integrate both sides)

Chapter 11 Parametric Equations and Polar Coordinates

Note 11.1 Parametrizations of Plane Curves

Definition of parametric Equations($x = f(t), y = g(t)$)

Parametric equations of the cycloid($x = a(t - \sin t), y = a(1 - \cos t)$ where a is radius and t is angle)

Note 11.2 Calculus with Parametric Curves

Parametric Formula for dy/dx ($\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$)

Parametric formula for d^2y/dx^2 ($\frac{d^2y}{dx^2} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$)(substitute y as y')

Length of a parametrically defined curve($L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt$)

Areas of Surfaces of revolution in parametrized curve

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \text{ (or } x \text{)}$$

Note 11.3 Polar Coordinates

Definition of polar coordinates (note for the rules of r)

Equations relating polar and Cartesian coordinates

Note 11.5 Areas and Lengths in Polar Coordinates

Area of fan-shaped region between origin and $r = f(\theta)$

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta$$

Area of the region between two polar curve ($A = \int_{\alpha}^{\beta} \frac{1}{2} (r_2^2 - r_1^2) d\theta$)

Length of a Polar Curve ($L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$)

Chapter 12 Vectors and the Geometry of Space

Note 12.1 Three-Dimensional Coordinate Systems

Basic concepts of space

Distance between two points in space

Standard equation for the sphere in space

Note 12.2 Vectors

Definition of vectors

Definition of component form for vectors

Definition of magnitude (length) of a vector

Definition of addition and scalar multiplication among vectors

Properties of vector operations

Definition of unit vectors and its information of direction

Formula of midpoint of a line segment

Note 12.3 The Dot Product

Definition of dot product and two formula

Dot product of perpendicular(orthogonal) vectors

Properties of the dot product

Scalar component and vector projection $\left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}$ where $\mathbf{u}$ onto $\mathbf{v}$

Note 12.4 The Cross Product

Definition of cross product $(\mathbf{u} \times \mathbf{v} = (|\mathbf{u}| |\mathbf{v}| \sin \theta) \mathbf{n})$

Cross product of parallel vectors

Properties of the cross product(note for

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w})$$

Geometric representation of $|\mathbf{u} \times \mathbf{v}|$ (the area of a parallelogram)

Determinant formula for cross product

Triple scalar product(box scalar product)($|\mathbf{u} \times \mathbf{v} \cdot \mathbf{w}|$) and its geometric representation(volume of a parallelepiped)

Calculating the triple scalar product as a determinant
$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

$$\begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Note 12.5 Lines and Planes in Space

Vector equation for a line ($\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, -\infty < t < \infty$)

Parametric Equations for a line

$$x = x_0 + tv_1, y = y_0 + tv_2, z = z_0 + tv_3, -\infty < t < \infty$$

Distance from a point to a line through P parallel to $\mathbf{v}$ ($d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$)

Equation for a plane in space with a point P_0 and normal vector (vector equations, component equations)

Distance from a point to a plane ($d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$ where $\mathbf{n}$ is normal vector to the plane)

Angle between planes ($\theta = \cos^{-1} \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$)

Chapter 13 Vector-Valued Functions and Motion in Space

Note 13.1 Curves in Space and Their Tangents

Definition of position vector function with respect to time

Limit and continuity of vector function

Derivative of position vector and velocity vector

Differentiable rules of vector function

Vector functions of constant length ($\mathbf{r}(t) = c \implies \mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0$)

Note 13.2 Integrals of Vector Functions; Projectile Motion

Definition of indefinite integral of vector functions in vector form and component form

Definition of definite integral of vector functions

Note 13.3 Arc Length in Space

Definition of the arc length along the position vector function(

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_a^b |\mathbf{v}| dt)$$

Arc length parameter with base point $P(t_0)$ (

$$s(t) = \int_{t_0}^t |\mathbf{v}(c)| dc = \int_{t_0}^t \sqrt{[x'(c)]^2 + [y'(c)]^2 + [z'(c)]^2} dc)$$

Speed vector of position function vector($\frac{ds}{dt} = |\mathbf{v}(t)|$)

Unit tangent vector($\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{d\mathbf{r}}{ds}$)

Note 13.4 Curvature and Normal Vector of a Curve

Definition of curvature and its information($\kappa = \left| \frac{d\mathbf{T}}{ds} \right|$)(degree of $\mathbf{T}$ changed with respect to s rather than t)

Formula for calculating curvature($\kappa = \left| \frac{d^2\mathbf{r}(s)}{ds^2} \right| = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right|$)

Definition of principal unit normal and its information($\mathbf{N} = \frac{1}{\kappa} \frac{d\mathbf{T}}{ds}$)(The direction information of κ)

Formula for Calculating $\mathbf{N}$ ($\mathbf{N} = \frac{\frac{d\mathbf{T}}{dt}}{\left| \frac{d\mathbf{T}}{dt} \right|}$)

Circle of curvature for plane curves and its features(have the same tangent line at that point; have the same curvature at that point)

Chapter 14 Partial Derivatives

Note 14.1 Functions of Several Variables

Basic concepts of several-variable function

domain and range, independent variables and dependent variables, interior point and boundary point, open and closed, bounded and unbounded.

Methods to graph several-variable function

- label curves(level curve)
- sketch the surface in space

Note 14.2 Limits and Continuity in Higher Dimensions

$\delta - \epsilon$ **Definition of limit for functions of more variables**

Properties of limits of functions of more variables

similar to single-variable function.

Definition of continuity

Two-Path test for **nonexistence of a limit**

Continuity of composites

Note 14.3 Partial Derivatives

Definition of partial derivatives

Methods of calculation

Relationship between derivatives and continuity(have partial derivatives not necessary continuity)

Higher-Order partial derivatives

The Mixed Derivative Theorem

Note 14.4 The Chain Rule

Chain rule for several-variable functions

Note 14.5 Direction Derivatives and Gradients Vectors

Definition of direction derivatives

Gradients and the dot product expression of direction derivatives(

$$\left(\frac{df}{ds} \right)_{\mathbf{u}, P_0} = (\nabla f)_{P_0} \cdot \mathbf{u}$$

Properties of the directional derivative and gradient

Relationship among the directions of gradient, tangent and level curve

Algebra Rules for Gradients

Note 14.6 Tangent Planes and Differentials

Definition and equations of tangent planes and normal lines(

$$\frac{\partial f}{\partial x}(x - x_0) + \frac{\partial f}{\partial y}(y - y_0) + \frac{\partial f}{\partial z}(z - z_0) = 0)$$

$$x = x_0 + \frac{\partial f}{\partial x}t, y = y_0 + \frac{\partial f}{\partial y}t, z = z_0 + \frac{\partial f}{\partial z}t)$$

Estimating change of the function in a specific direction(

$$df = (\nabla f|_{P_0} \cdot \mathbf{u})ds)$$

Linearization of several-variable function(

$$L(x, y) = f(x_0, y_0) + \frac{\partial f}{\partial x}(x - x_0) + \frac{\partial f}{\partial y}(y - y_0))$$

Definition of total differential($df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$)

Note 14.7 Extreme Values and Saddle Points

Definition of local extreme values and first derivative test for local extreme values

Definition of critical point

Definition of saddle point

Second derivative test for local extreme values

	f_{xx}	$f_{xx}f_{yy} - f_{xy}^2$ ($\begin{vmatrix} < br > f_{xx} & f_{xy} \\ < br > f_{xy} & f_{yy} \\ < br > \end{vmatrix}$)
local maximum	< 0	> 0
local minimum	> 0	> 0

	f_{xx}	$f_{xx}f_{yy} - f_{xy}^2$ $($ $\left\ \begin{matrix} < br > f_{xx} & f_{xy} \\ < br > f_{xy} & f_{yy} \end{matrix} \right\ $ $< br >$ $)$
saddle point	\	< 0
inconclusive	\	$= 0$

Methods to find absolute maxima and minima on closed bounded regions

Note 14.8 Lagrange Multipliers

Usage of Lagrange multiplier(finding extreme values of constrained functions)

The detailed method of Lagrange multipliers($\nabla f = \lambda \nabla g, g = 0$)

Lagrange multipliers with two constraints(

$\nabla f = \lambda \nabla g_1 + \mu \nabla g_2, g_1 = 0, g_2 = 0$)

Chapter 15 Multiple Integrals

Note 15.1 Double and Iterated Integrals over Rectangles

Definition and understanding of double integrals

Fubini's Theorem for calculating double integrals(

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx)$$

Note 15.2 Double Integrals over General Regions

Riemann sum over general regions

Stronger form of Fubini's Theorem(

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx)$$

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(x)}^{h_2(x)} f(x, y) dx dy$$

Properties of double integrals

Note 15.3 Area by Double Integration

Definition of area and formula in double integration ($A = \iint_R dA$)

Average value of double integrals ($A_{\text{ve}} = \frac{1}{A} \iint_R f dA$)

Note 15.4 Double Integrals in Polar Form

Integrals in polar coordinates ($\iint_R f(r, \theta) dA = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=g_1(\theta)}^{r=g_2(\theta)} f(r, \theta) r dr d\theta$)

Area in polar coordinates ($A = \iint_R r dr d\theta$)

Changing Cartesian integral into polar integrals ($\iint_R f(x, y) dx dy = \iint_G f(r \cos \theta, r \sin \theta) r dr d\theta$)

Note 15.5 Triple Integrals in Rectangular Coordinates

Definition and understanding of triple integrals and Fubini's theorem on triple integral

Volume of a region in space in triple integration ($V = \iiint_D dV = \iiint_D dx dy dz$)

Average value of triple integrals ($a_{\text{ve}} = \frac{1}{D} \iiint_D F dV$)

Properties of triple integrals

Note 15.6 Moments and Centers of Mass

Mass expressed by triple integrals of density ($M = \iiint_D \delta(x, y, z) dV$)

Understanding and formula of First moment ($M_r = \iiint_D r \delta dV$)

Center of mass and its computation

Centroid obtained by constant density

Moment of inertia(second moment) and its formula(

$$I = \int r^2 dm = \int r^2 \delta dV)$$

Note 15.7 Triple Integrals in Cylindrical and Spherical Coordinates

Definition of cylindrical coordinates and its equations(

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z \quad r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x})$$

$$\text{Integrals of cylindrical coordinates}(\iiint_D f r dr d\theta dz = \iiint_D f dz r dr d\theta)$$

Definition of spherical coordinates and its equations(

$$r = \rho \sin \phi, x = r \cos \theta = \rho \sin \phi \cos \theta \quad z = \rho \cos \phi, y = r \sin \theta = \rho \sin \phi \sin \theta \quad \rho)$$

$$\text{Integrals of spherical coordinates}(\iiint_D f(\rho, \phi, \theta) \rho^2 \sin \phi d\rho d\phi d\theta)$$